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PFD473S: Term 1 Project Management Concepts

Introduction to Project Management (SO1)

What is a Project?

A project, as defined by the PMBOK Guide, is a **temporary endeavor** undertaken to create a **unique product, service, or result**. This distinguishes it from operations, which are ongoing and repetitive.

Key characteristics of a project include:

- **Temporary:** Every project has a defined beginning and end. The end is reached when objectives are achieved, or it's clear they cannot be met. This does not imply a short duration, only a finite one.
- **Unique:** Projects create distinct deliverables, whether a tangible product, a new service capability, or a research finding.
- **Progressive Elaboration:** This is the characteristic of developing a project in steps and increments. The scope is broad initially and becomes more detailed over time as understanding grows. This is a planned process and should not be confused with scope creep.

Projects share commonalities with operations, such as being performed by people, being constrained by limited resources, and requiring planning, execution, and control.

The Iron Triangle (Project Constraints)

The Iron Triangle illustrates the fundamental trade-offs between three primary project constraints:

- Scope
- Time
- Cost

Quality is often considered the central constraint, influenced by changes in the other three. Adjusting one constraint inevitably affects the others.

Project Management Definition

Project management is the application of knowledge, skills, tools, and techniques to project activities to meet project requirements. It is accomplished through the processes of initiating, planning, executing, monitoring and controlling, and closing.

An alternative functional view outlines four basic activities:

- **Planning:** Specifying results, activities, schedules, and resources.
- **Organizing:** Defining the project structure, roles, and responsibilities, mapping roles to work.
- **Controlling:** Identifying deviations from the plan, reallocating resources, and adjusting schedules.
- **Terminating:** Delivering to the client, reviewing lessons learned, and updating templates.

The Project Manager's Role: Project managers are responsible for identifying requirements, establishing clear objectives, balancing competing demands (scope, time, cost, quality), and adapting plans for stakeholders. They often have high responsibility but limited formal authority, relying on influence to achieve project goals.

Organizational Types

- **Functional (Line) Organisation:** Participants are grouped by specialty into permanent departments. Projects move through departments. This is a fixed structure, with few

duration. This structure is dynamic, with projects created on request and terminated upon delivery.

- **Matrix Organisation:** This type combines functional and project-based structures. Participants work in both their department and on projects simultaneously. Each member reports to both a department head and a project manager. This structure allows for rapid response and skill recruitment but violates Fayol's principle of "Unity of Command."

Project Management Body of Knowledge (PMBOK) Knowledge Areas

The PMBOK Guide identifies nine key knowledge areas for project management:

- Project Integration Management
- Project Scope Management
- Project Time Management
- Project Cost Management
- Project Quality Management
- Project Human Resource Management
- Project Communications Management
- Project Risk Management
- Project Procurement Management

Work Breakdown Structure (WBS)

The WBS is a hierarchical representation of all tasks within a project. It serves as the foundation for scheduling, resource allocation, and cost estimation.

- **Task:** The smallest unit of work subject to project management.
- **Work Package:** Describes the work products to be produced, resources needed, expected duration, dependencies, and acceptance criteria.
- **Deliverable:** Any work product delivered to the customer.
- **Internal Work Product:** Work products consumed only by project participants.

Management Theory (SO2)

Henri Fayol's Management Principles

Fayol outlined five functions of management:

- **Prevoyant (Forecast & Plan):** Examining the future and drawing up a plan of action.
- **Organise:** Building the material and human structure of the undertaking.
- **Command:** Maintaining activity among personnel.
- **Coordinate:** Unifying and harmonizing all activity and effort.
- **Control:** Ensuring conformity with established rules and commands.

Fayol also proposed 14 Principles of Management:

1. **Specialisation of labour:** Encourages skill improvement and method development.
2. **Authority:** The right to give orders and exact obedience.
3. **Discipline:** Adherence to rules and absence of slacking.
4. **Unity of Command:** Each employee should have one and only one boss. (Violated in matrix organizations).
5. **Unity of Direction:** A single plan generated by one mind, with all playing their part.
6. **Subordination of Individual Interests:** Work interests should take precedence over personal interests.
7. **Remuneration:** Employees should receive fair payment for their services.
8. **Centralisation:** Consolidation of management functions, with decisions often made from the top.
9. **Scalar Chain (Line of Authority):** A formal chain of command from top to bottom.

12. **Personnel Tenure:** Limited turnover of personnel for stability and efficiency.

13. **Initiative:** Encouraging employees to think out and execute plans.

14. **Esprit de Corps:** Fostering harmony and cohesion among personnel.

Distinction: Do not confuse **Remuneration** (fair payment) with **Equity** (equality of treatment).

Henry Mintzberg's Managerial Roles

Mintzberg studied what managers actually do, identifying 10 roles categorized into three groups. These roles are inseparable and form a gestalt.

Interpersonal Roles:

- **Figurehead:** Performing ceremonial duties by virtue of position.
- **Leader:** Responsible for the work of others, motivating and encouraging the team. Project managers often rely heavily on influence in this role due to limited formal authority.
- **Liaison:** Making contacts outside the vertical chain of command to build external information networks.

Informational Roles (approximately 40% of a manager's time):

- **Monitor:** Constantly scanning the environment for information, including informal "soft" information like gossip and hearsay, to anticipate problems.
- **Disseminator:** Distributing relevant information to the team. Hoarding information hinders team decision-making.
- **Spokesperson:** Passing information to people outside the manager's unit (e.g., to upper management or clients).

Decisional Roles:

- **Entrepreneur:** Initiating development projects when a good idea is identified. This is a proactive role.
- **Disturbance Handler:** Reacting to pressures and changes outside the manager's control (e.g., scope changes, accidents, personnel loss). This is a reactive role.
- **Resource Allocator:** Deciding who gets what resources, including the manager's own time.
- **Negotiator:** Engaging in negotiations with stakeholders, vendors, or other parties.

Distinctions:

- **Entrepreneur vs. Disturbance Handler:** Entrepreneur is proactive (manager initiates); Disturbance Handler is reactive (manager responds to external pressure).
- **Monitor vs. Disseminator:** Monitor gathers information; Disseminator shares information with the team.

PERT Calculation Example

PERT (Program Evaluation and Review Technique) is used for projects where activity durations are uncertain. It uses probabilistic time estimates.

Steps:

1. **Draw the Network:** Use Activity-on-Arrow (AOA) or Activity-on-Node (AON) conventions.
2. **Calculate Expected Durations (ED):** For each activity, estimate three durations:
 - Optimistic Time (A): The shortest possible time if everything goes perfectly.
 - Most Likely Time (M): The duration under normal conditions.
 - Pessimistic Time (B): The longest plausible duration if significant problems occur.

The Expected Duration is calculated using a weighted average, often assuming a Beta distribution:

$$ED = \frac{A + 4M + B}{6}$$

3. **Calculate variance:** The variance for each activity's duration is:

$$\text{Var} = \left(\frac{B - A}{6} \right)^2$$

4. **Forward Pass:** Calculate the Early Start (ES) and Early Finish (EF) times for each activity.

- $ES_{activity} = \max(EF_{predecessors})$
- $EF_{activity} = ES_{activity} + ED_{activity}$

5. **Backward Pass:** Calculate the Late Start (LS) and Late Finish (LF) times for each activity.

- $LF_{activity} = \min(LS_{successors})$
- $LS_{activity} = LF_{activity} - ED_{activity}$

6. **Identify the Critical Path:** The critical path consists of activities with zero slack (float). Slack is the difference between an activity's latest and earliest times ($LS - ES$ or $LF - EF$).

7. **Calculate Total Variance for the Project:** Sum the variances of all activities on the critical path. Let this sum be S^2 .

$$S^2 = \sum(\text{Var}_{\text{critical path activities}})$$

The standard deviation of the project duration is $S = \sqrt{S^2}$.

8. **Calculate Probability:** To find the probability of completing the project by a specific target date (x), use the Z-score formula, assuming a normal distribution for the project completion time:

$$z = \frac{x - \mu}{S}$$

Where μ is the expected project completion time (usually the EF of the last activity on the critical path). Look up the calculated z -value in a standard normal cumulative probability table to find the probability. Alternatively, if a probability is given and you need to find the completion time (x):

- Look up the z -value corresponding to the probability in the table.
- Calculate the time using:

$$x = zS + \mu$$



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